



Assessing crayfish, *Procambarus clarkii*, vision through looming stimuli associated with anti-predator behaviors

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INTRODUCTION

- Crayfish exhibit anti-predator behaviors in response to predators, such as escape responses like tail-flipping or walking away.¹
- Previous studies have shown that crayfish primarily respond through chemical and tactile cues, with visual cues playing a secondary role.^{2,3}
- Crayfish also exhibit neutral behaviors (e.g., freezing behavior).⁴
- Visual cues, like looming stimuli, can trigger defensive behaviors in mantis shrimp, closely related to crayfish.⁵
- Our experiment investigates where a looming stimulus—a visual cue resembling an approaching predator—elicits escape behaviors (e.g., tail-flipping) in crayfish.

AIMS

- **RQ1:** Will crayfish show anti-predator behaviors when they see a looming stimulus approaching the tank?
- **H1:** If a crayfish sees a looming stimulus approaching its frontal vision, then the crayfish will elicit an escape behavior rather than a neutral behavior.

METHODS & MATERIALS

- 11 crayfish (fig. 2) were individually tested 4 times: two trials with looming and 2 trials with receding stimulus of a bass (fig. 4)
- Each crayfish was given 30 minutes to acclimate before testing their vision in front and on their right peripheral vision (fig. 1)
- The statistical test, chi-squared, was conducted.

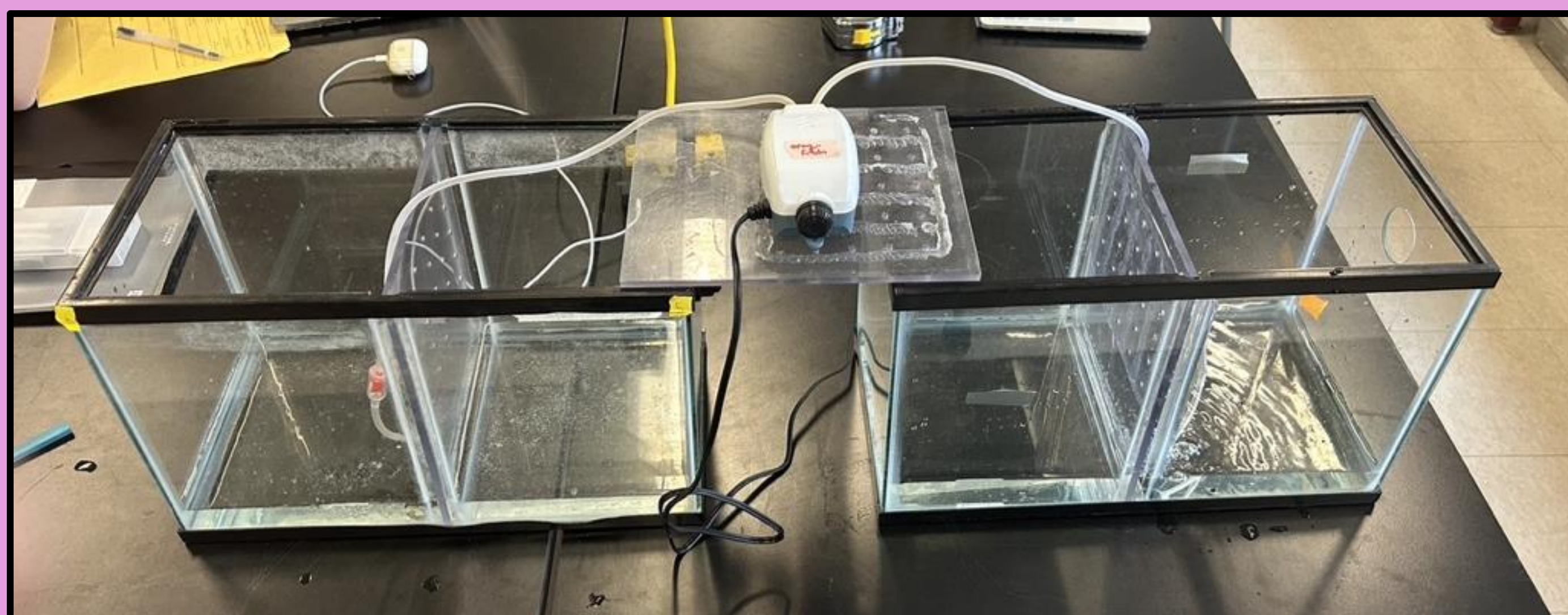


Figure 1. Set up of the Experiment

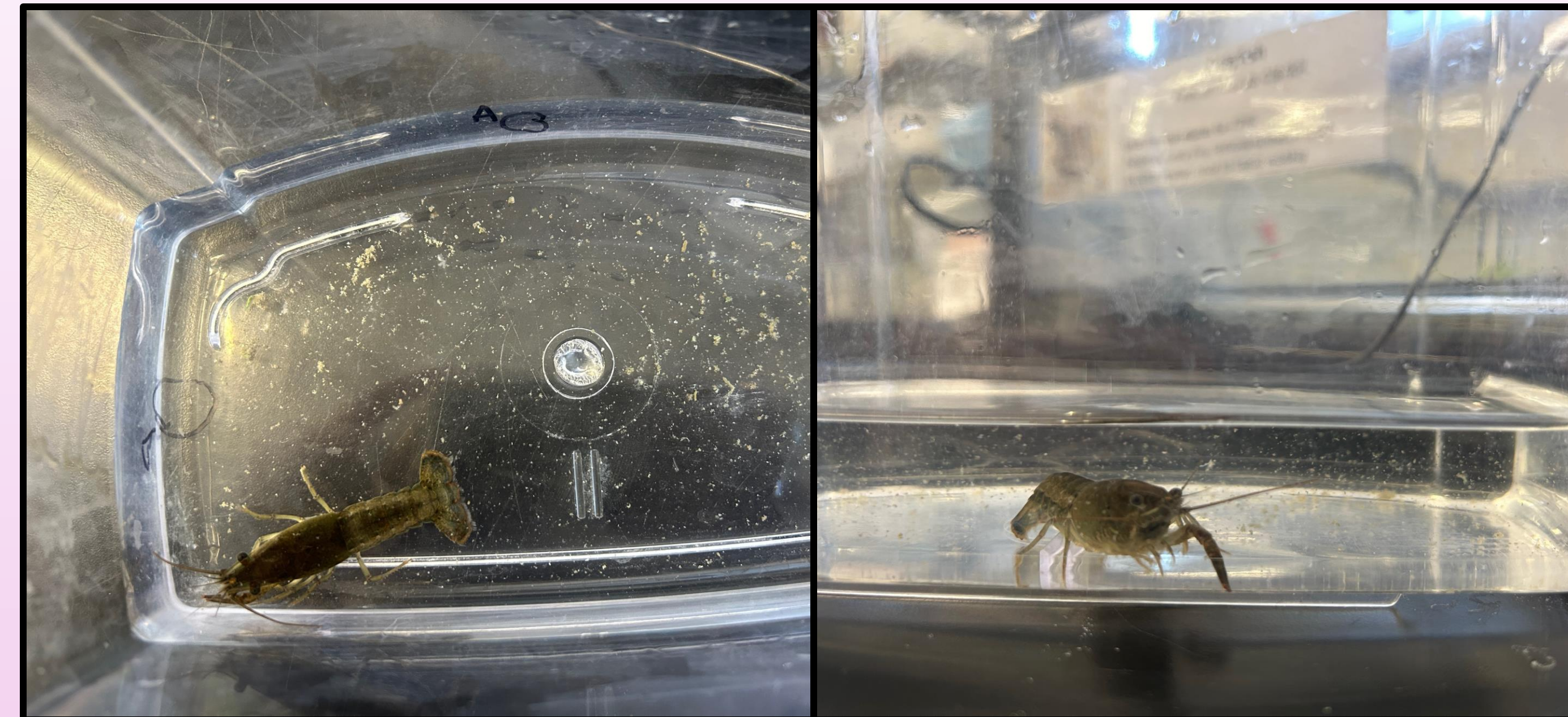


Figure 2: Image of an experimental subject, *Procambarus clarkii*

RESULTS

- Figure 3 displays the activity of crayfish in response to a stimuli
- Crayfish did not have a significant active response when shown a looming stimulus (chi-square test of independence, $\chi^2 = 0.08$; $P = 0.773$)
- There were no differences in behavior when shown a looming or receding stimuli between trials

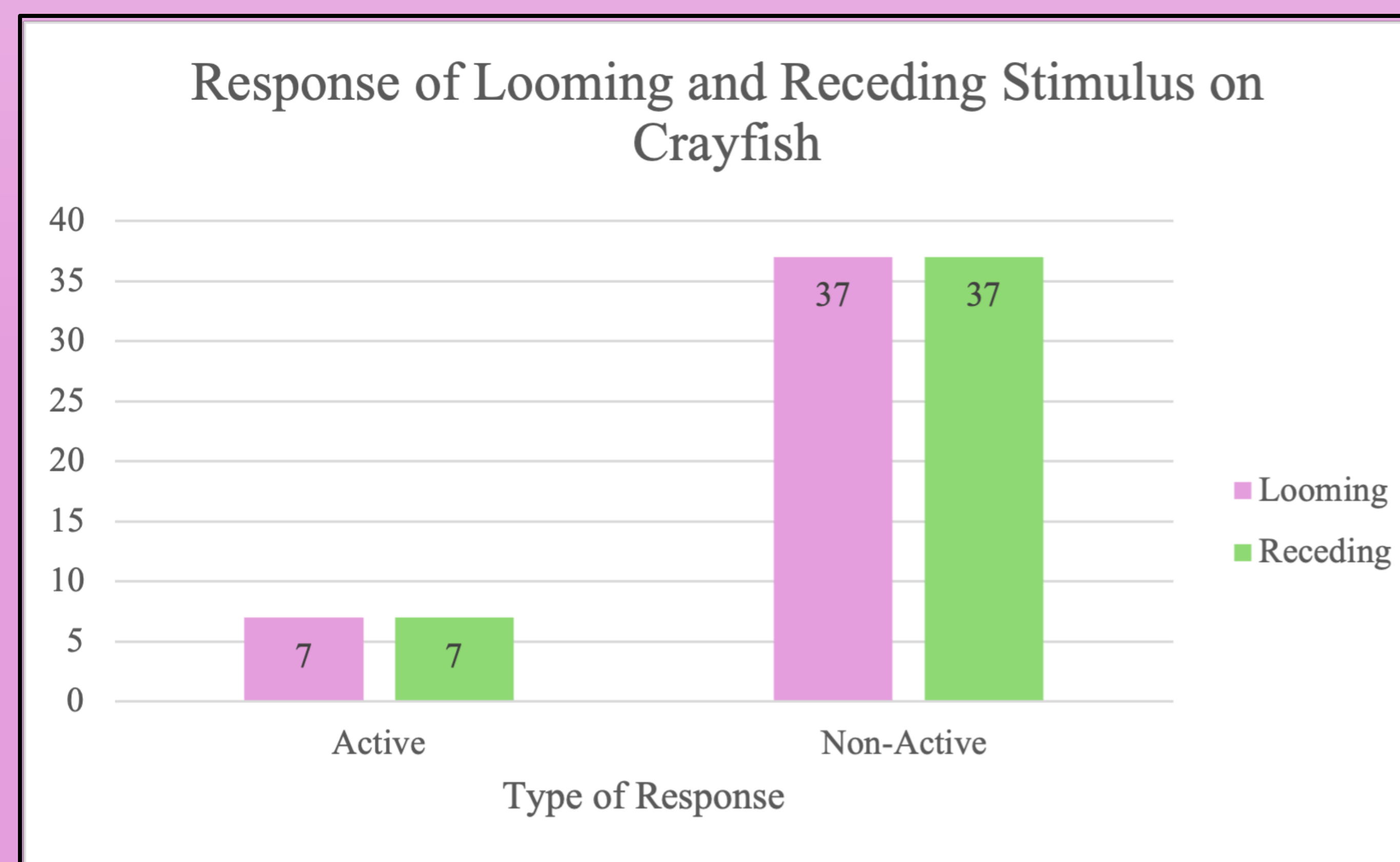


Figure 3. Responses of Looming and Receding Stimuli based on both trials. Active responses include tail-flip and walk away while non-active includes stationary.

DISCUSSION

- Based on the statistical test, the null hypothesis is unable to be rejected.
- As shown by the results, there is no significant evidence to prove that our visual stimuli alone can elicit an active response.
- The use of laboratory-raised crayfish may have altered the results.
 - Crayfish respond to stimuli based on learned experiences.² The lack of predator threats while they were raised may have caused a lack of active anti-predator behaviors.
- In addition to vision, crayfish may need their olfactory and tactile senses
 - Crayfish will tail-flip when multiple senses stimulate them.⁴
 - There were instances of crayfish readjusting towards the screen when a looming stimulus occurred.
 - Vision needs to be paired with other senses to show that a threat is present.
- Future studies should study the visual, tactile, and olfactory senses simultaneously.
 - Focus on wild crayfish with learned behaviors and laboratory-raised with no predator experience to see differences with learned experiences and senses



Photo by S. Neuharth

Figure 4. Image used for Looming and Receding Stimuli During Trials

References

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